Md Saif Rahman

Dhaka,Bangladesh | rahmansaif.413@gmail.com | +880 1763 742475 | My Portfolio | LinkedIn | Github

Professional Summary

Prospective PhD applicant with research interests in natural language processing, large language models, and trustworthy AI. My research experience spans multilingual text analysis, transformer-based mental health detection, sentiment analysis, and Bangla–Banglish NLP. I am interested in developing robust, inclusive, and linguistically adaptable AI systems, with broader applications in healthcare and human-centered AI.

Education

Bangladesh University of Professionals, B.Sc. in Information and Communication Engineering — 2020 – August 2025

- GPA: 3.63/4.0
- **Coursework:** Object-Oriented Programming, Analysis and Design of Algorithms, Computer Networking, Machine Learning, Artificial Intelligence

Dr. Mahbubur Rahman Molla College, Higher Secondary Certificate (H.S.C.) — 2018 – 2020

- GPA: 5.0/5.0

Publications

Saif Rahman, Tarannum Binte Tariq, Nusrot Jahan, Naimul Hasan Naime, Nazneen Akhter
HAN-DistilBERT for Depression Detection from Twitter Data: Integrating Hierarchical Attention with Transformer-Based Model
International Conference on Computer and Information Technology (ICCIT) 2024
DOI: 10.1109/ICCIT64611.2024.11022022

Tarannum Binte Tariq, Nusrot Jahan, ***Saif Rahman***, Naimul Hasan Naime (Accepted)
Enhanced Twitter-Based Depression Severity Classification Using Multi-Head Attention with RoBERTa and DistilBERT Fusion
IEEE 7th International Conference on Sustainable Technologies for Industry 5.0 (STI 2025)

Saif Rahman, Tarannum Binte Tariq, Naimul Hasan Naime, Nazneen Akhter, Dr Md Sazzadur Rahman (On review)
Linguistic Feature Based Depression Detection in Social Media Across Bangla, Banglish, and English
Expert Systems and Applications

Research Interests

Artificial intelligence and machine learning; natural language processing; multilingual and low-resource language processing; transformer-based models; computational mental-health research; and responsible AI.

Skills

Machine Learning: Scikit-learn, regression, classification, feature engineering, model evaluation, data preprocessing
Deep Learning: TensorFlow, Keras, CNNs, neural networks, transfer learning
AI/NLP: Hugging Face Transformers, BERT, RoBERTa, DistilBERT, text classification
Programming: Python, C++, JavaScript
Data Analysis: Pandas, NumPy, Matplotlib, Seaborn, exploratory data analysis (EDA)
AI Engineering: LangGraph, LangChain, LLM integration, AI agents, prompt engineering
Backend and Deployment: FastAPI, REST APIs, Streamlit, model deployment
Databases: MySQL, MongoDB
Version Control: Git, GitHub

Professional Experience (Industrial Experience)

Junior Software Engineer, Void Studio July 2024 – August 2025

- Developed a full-featured e-commerce website for a renowned bookshop company, focusing on front-end development and UI design.
- Improved website performance and loading speed through front-end optimization techniques.
- Designed and developed the company's portfolio website to enhance its online presence and branding.
- Contributed as a front-end developer to a medical admissions platform, focusing on user experience and interface design.
- Collaborated on multiple web development projects, gaining experience in responsive design and modern technologies.

IT Intern, Trust Bank PLC December 2024 – January 2025

- Led a frontend team to develop a fully responsive banking CMS.
- Conducted research on national and international banking platforms to tailor the design to user needs.
- Created Figma prototypes and implemented them using HTML, CSS, Bootstrap, JavaScript, and PHP.
- Enhanced mobile and desktop experiences through performance optimization and intuitive navigation.
- Simplified a complex multi-page CMS into a clean, user-centric platform, improving accessibility and usability.

Projects

Diabetes Prediction using Machine Learning GitHub Link

- Developed a machine learning-based diabetes prediction system through data preprocessing, exploratory data analysis, and feature engineering.
- Handled missing values and outliers and applied data transformation techniques to improve model performance.
- Trained and evaluated multiple ML models using appropriate performance metrics.
- Tools Used: Python, Scikit-learn, Pandas, NumPy, Matplotlib

ACI Product Demand Predictor GitHub Link

- Built a machine learning-based demand forecasting system using historical sales data.
- Performed data analysis, feature engineering, and model comparison to identify the best-performing regression model.
- Implemented and evaluated LightGBM, XGBoost, and Linear Regression.
- Tools Used: Python, LightGBM, XGBoost, Scikit-learn, Pandas

Potato Disease Classification using CNN GitHub Link

- Developed a deep learning-based image classification model to detect potato leaf diseases.
- Applied image preprocessing, data augmentation, and a CNN-based architecture for disease classification.
- Trained and evaluated the model to automate agricultural disease detection.
- Tools Used: Python, TensorFlow, Keras, CNN, OpenCV

GroqFlow - AI Agent System GitHub Link

- Developed an AI agent application using LangGraph to create intelligent workflow-based automation.
- Built a full-stack AI system with a FastAPI backend and Streamlit frontend.
- Integrated LLM capabilities using Groq Cloud, LLaMA models, and the Tavily search API.
- Tools Used: Python, FastAPI, Streamlit, LangGraph, Groq Cloud, LLaMA

Nilkhet Boighor Website Link

- Refactored the frontend architecture by migrating the project from React.js to Next.js for improved scalability and performance.
- Redesigned and optimized the UI/UX to deliver a faster, cleaner, and more user-friendly experience.
- Fixed frontend bugs and improved application stability and responsiveness.

- Converted large frontend modules into reusable and maintainable React components.
- Tools Used: Next.js, React.js, Tailwind CSS

GRE Score

Quantitative	Verbal	Analytical Writing (AWA)
163	154	3.5

Volunteering and Leadership Experience

Vice Chair, IEEE Computer Society BUP Student Branch Chapter May 2024 – August 2025

- Contributed to logistics and graphics teams, ensuring smooth execution of events and activities.
- Designed promotional posters and visual materials for an international conference.
- Assisted in organizing interviews, viva sessions, and technical evaluations.

Club Coordinator, BUP Robotics Club May 2024 – August 2025

- Led frontend development of the club's official website, focusing on UI/UX design and responsiveness.
- Collaborated with team members to build and maintain the club's online presence.
- Participated in workshops and technical events to enhance practical knowledge.

Joint Secretary, BUP InfoTech Club April 2024 – August 2025

- Designed social media posters and promotional content for club events and activities.
- Assisted in organizing events, workshops, and meetings, ensuring smooth coordination.
- Supported interview processes and contributed to overall club management operations.